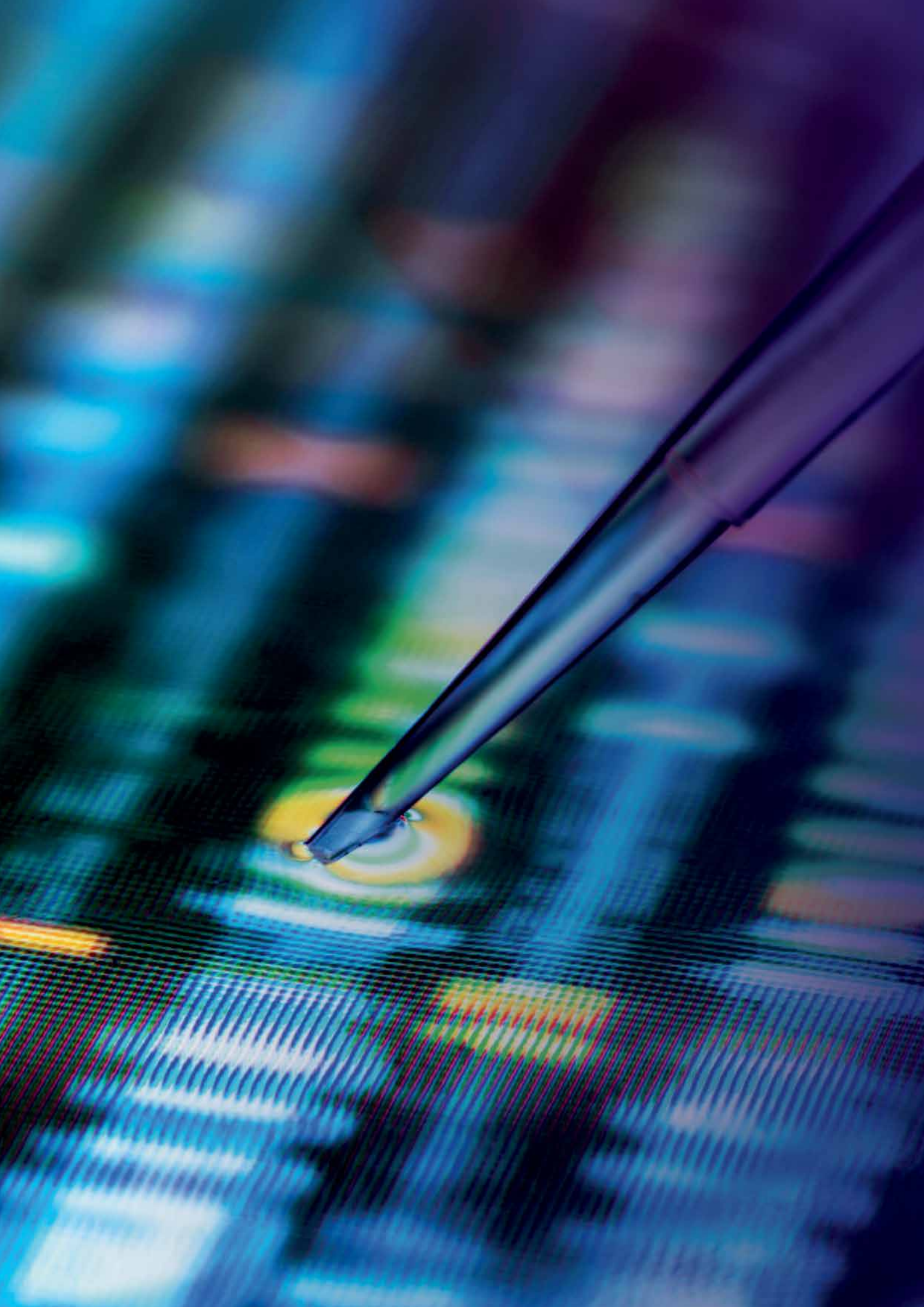


Pathogen genomic sequence data sharing

Exploring technical solutions
for low-resource settings





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1. Executive summary

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1. Executive summary

1.1 Aims and scope

Pathogen genomic sequence (PGS) data sharing is crucial for responding to public health challenges, as evidenced by the COVID-19 pandemic. However, PGS data generators and users in low-resource settings face significant barriers to PGS data sharing due to gaps in infrastructure and expertise, and inequitable benefits from sharing these data. Wellcome worked with global stakeholders to understand these challenges and explore technical solutions that have the potential to catalyse and improve how these data are shared. The insights in this report were gathered through a multi-stage process, involving a survey, interviews and two workshops held in October 2024.

1.2 Technical solutions to enhance PGS sharing

The term ‘low-resource settings’ is used in this report to refer to the availability of resources and expertise, which will vary between and within national contexts and is not geographically bound. This work considered a wide range of technical solutions currently in use to assemble, annotate, analyse and share PGS data in these contexts. These solutions enable PGS data generators and users in low-resource settings to access global expertise, tools and networks but mean they are frequently dependent on external resources. Data generators’ willingness and ability to share data is also constrained by uncertainty over what data should be shared, and in what context, and concerns over a lack of recognition and credit.

Twelve areas of opportunity to enhance existing technical solutions emerged from this work, which were prioritised as shown in the figure below. However, contributors emphasised that data sharing also relies on an enabling environment characterised by trusted relationships, effective data governance and policy, robust systems for recognition and credit, and education and advocacy efforts.

1.2.1 Prioritisation of technical solutions for PGS data sharing

	Sharing PGS data	Analysing PGS data	Reusing PGS data
High priority	- Improved data integration and interoperability - User-friendly data upload and sharing systems	- Open-source bioinformatics tools that enable local interpretation - Collectively agreed-upon metadata standards	- Mechanisms for recognition and credit - Frameworks for ‘minimum data sharing’
Medium priority	Standard data formats and protocols for uploading	Support for federated analysis without data transfer	Tracking reuse and impact of datasets
Low priority	Low bandwidth solutions for data sharing	Improved tools for data visualisation	Systems for managing data ownership and governance

1.3 How funders can help

The workshops showed that funders like Wellcome are in a strong position to drive improvements in global sharing of PGS data. Participants highlighted the potential to **enhance technical infrastructure** for data upload, visualisation and analysis, but stressed that interventions in this area must be accompanied by efforts to **strengthen local capacity** and **build community norms**. The interconnected nature of these activities points to a need for coordination between funding agencies to minimise duplication and maximise their collective impact.



1.3.1 How funders can support PGS data sharing in low-resource settings

	Enhancing technical infrastructure	Strengthening local capacity	Building community norms
What funders can do	Support development of user-friendly data upload systems suitable for low-bandwidth contexts and interoperable with international systems.	Embed training in the development of new solutions, following a 'train the trainer' model for sustainable knowledge dissemination.	Promote adoption of globally applicable minimum standards and protocols for data collection and sharing.
	Support development of open source, federated data visualisation and analysis tools.	Tailor capacity-strengthening efforts to address both technical and contextual challenges.	Promote best practices for data sharing and recognition.

Cross-cutting themes

Build on existing solutions and networks

To advance PGS data sharing, it is crucial to build on existing global platforms, COVID-19 momentum, and learn from other sectors. Supporting services and data sharing platforms that implement transparent governance and respect data generators' rights to control usage are essential. These approaches can foster trust and efficient collaboration.

Deliver through transdisciplinary teams

Teams addressing PGS data sharing should integrate legal, technical, governance, public health and policy expertise. Solutions should balance data generator and user interests, while incorporating advocacy and capacity-building expertise to support implementation and adoption of new technical approaches.

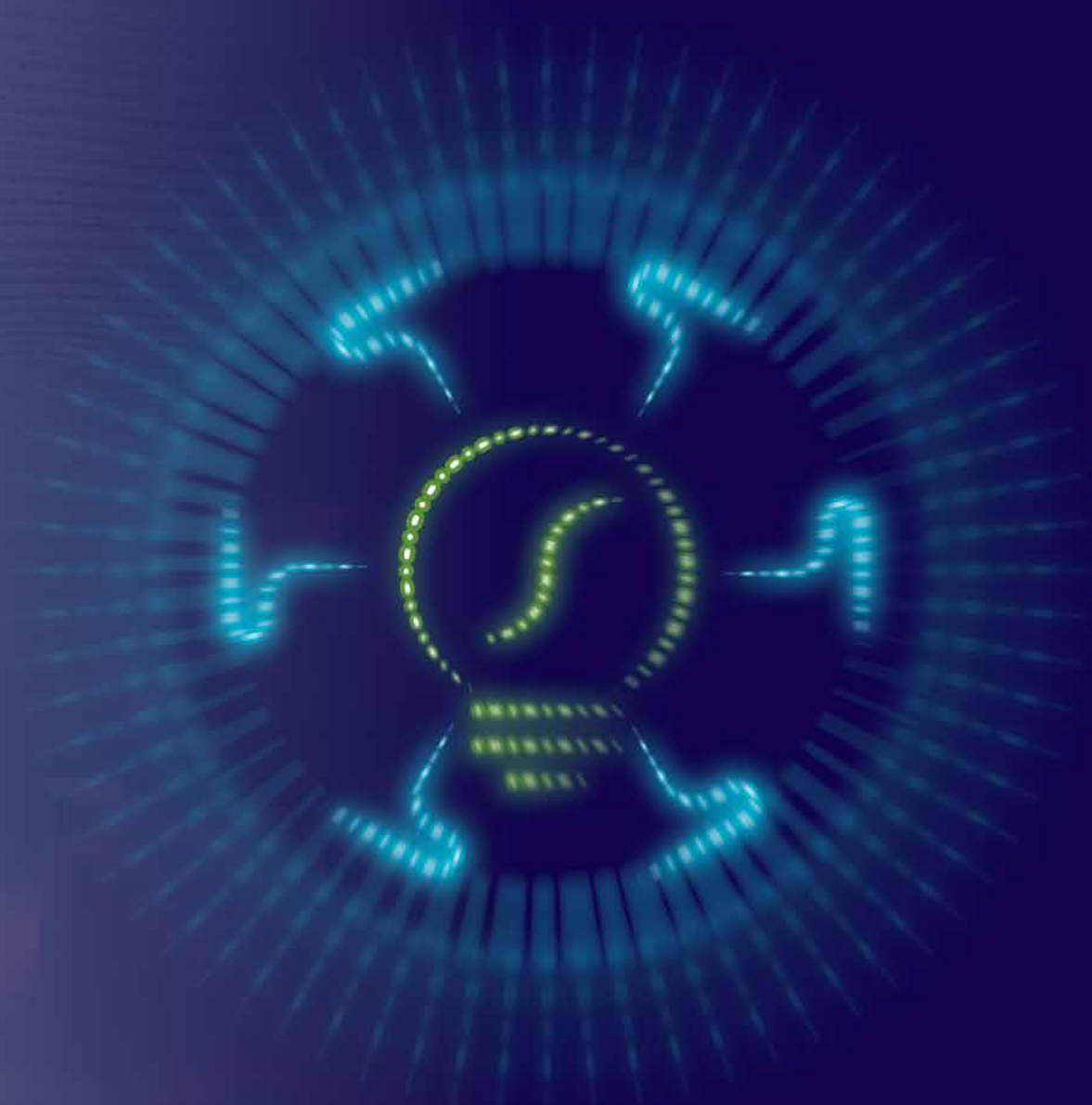
Adapt to local contexts and user needs

Successful data sharing solutions must be co-developed with local communities, ensuring interoperability while addressing regional variations in infrastructure and needs. Flexible and transparent governance and innovative approaches should accommodate low-resource settings, bridge fragmented systems, and adapt to different data types and sensitivity levels across research and public health contexts.



2. Introduction

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2. Introduction

2.1 Rationale

Sharing pathogen genomic sequence (PGS) data was [crucial](#) during the COVID-19 pandemic, supporting disease surveillance, tracking variants of concern, and aiding the development of diagnostics, therapeutics, and vaccines for SARS-CoV-2. Recognised as vital for understanding and responding to infectious diseases globally, data sharing has enabled researchers, policymakers, and public health bodies to act more quickly and knowledgeably. Beyond crises, the continuous sharing of PGS data plays an essential role in the routine monitoring and control of infectious diseases, contributing significantly to [public health preparedness](#), prevention strategies, and broader health initiatives. PGS data sharing has played a significant role in managing influenza, Ebola, Zika, and Mpox outbreaks.

However, stakeholders in low-resource settings continue to face difficulties both as data generators and as data users in sharing and accessing PGS data. While there are many cultural, ethical, social, legal, and economic challenges surrounding data sharing, one of the crucial issues is the technical difficulty and skills mismatch involved in enabling deposit, access and re-use within low-resource contexts. Wellcome held two virtual workshops to understand the current technical challenges facing people who interact with PGS data in low-resource settings and explore how funding can support data sharing. This report highlights the outcome of these workshops and the proposed solutions to meet the needs of those sharing PGS data in low-resource settings.

2.2 Report aims

The aims of the workshops were as follows:

- Understand the barriers faced by data generators and users when analysing, depositing, accessing and re-using PGS data in low-resource settings.
- Explore potential technical solutions that can overcome these barriers and enable the effective use of PGS data.
- Inform the strategic approach of Wellcome and other funders in this space, with a particular focus on Africa and Asia.

Based on analysis from both workshops, together with a preparatory survey and interviews, this report offers recommendations to Wellcome and other funders on the key priority areas within the technical space to improve PGS data sharing in low-resource settings.

2.3 Scope

A complex set of cultural, ethical, social, legal and economic factors make sharing, analysing and re-using PGS data challenging. However, the workshops were designed to examine the specific technical barriers that cause difficulties in low-resource settings, and the types of technical solutions that may mitigate these and support successful PGS data sharing.

For the purposes of the workshops, Wellcome adopted the following definition:

“Technical solutions or tools in the context of pathogen genomic sequence data sharing include physical or cloud infrastructures, tooling/software for assembling, annotating, or analysing sequence data, secure ‘Trusted Research Environments (TREs)’, data sharing and collaboration platforms, application programme interfaces (APIs) and middleware that facilitates automated data exchange, as well as simpler sharing mechanisms like email, physical hard drives and any other technical solution or software that is used for the purposes of sharing pathogen genomic sequence data.”

This report therefore focuses primarily on the technical solutions that funders could consider supporting in low-resource settings but also incorporates additional considerations based on feedback from workshop participants. These include broader aspects of data sharing beyond the technical scope which can help to inform a comprehensive strategy in this domain.



2.4 The wider context

Global actors such as the Public Health Alliance for Genomic Epidemiology (PHA4GE) [Working Group on data sharing and ethics](#), the [World Health Organization \(WHO\)](#), through [its guiding principles for PGS data sharing](#) and the work of the [International Pathogen Surveillance Network \(IPSN\)](#), are also building momentum in PGS data sharing to ensure equitable access to sequencing technologies and its benefits. Other notable regional-specific or country-specific consortia include, but are not limited to, the [Asia Pathogen Genomics Initiative](#), the [African Biogenome Project](#), [Africa Centres for Disease Control and Prevention](#), the [Pan African Bioinformatics Network for the Human Hereditary and Health in Africa \(H3ABioNet\)](#), and the [Indian SARS-CoV-2 Genomics Consortium \(INSACOG\)](#).

Cognisant of these other developments, Wellcome feels that it is particularly well-placed to champion PGS data sharing by supporting successful and workable technical solutions in low-resource settings.

2.5 Methodological approach

The term ‘low-resource settings’ is used in this report to refer to the availability of resources and expertise, which will vary both between and within national contexts. Even in high-income countries, some research and public health stakeholders may have limited resources for pathogen data sharing due to the level of specialised skills and technical solutions required.

The insights in this report were gathered through a multi-stage process. Initially, a global survey was conducted in July 2024, receiving 71 responses from stakeholders, with 63% of respondents from low- and middle-income countries. The survey aimed to identify the barriers and potential solutions for PGS data sharing. Following the survey, four in-depth interviews were conducted with experts to dive deeper into the technical challenges and possible solutions specific to low-resource environments. These findings were further explored in two virtual workshops, where participants discussed these issues in detail, collaborating to propose practical solutions for PGS sharing.

The thematic analysis and coding of the qualitative data for this report were supported by AI tools. [Claude](#), a large language model, was used to suggest themes, codes, and labels for the interview and workshop transcripts based on the context and objectives provided. The authors reviewed and refined Claude’s suggestions to ensure the validity and reliability of the analysis. Additionally, the drafting of certain sections of this report was assisted by [Microsoft CoPilot Pro](#), which facilitated the writing process. The combined use of these AI tools occurred in October 2024 and contributed to the effective preparation of this analytical report.

2.6 Acknowledgements

The project team at Research Consulting gratefully acknowledges the support of Julia Abernethy, Natsuko Imai-Eaton and Athena Michaelides, as well as the wider staff at Wellcome.

We would like to thank Meera Chand and ThankGod Ebenezer for their contributions as peer reviewers for this report. We would also like to thank Alan Christoffels, Duncan MacCannell, Nicky Mulder and ThankGod Ebenezer for engaging in interviews with Wellcome and the Research Consulting team.

Finally, we are deeply appreciative of all the participants who generously gave their time and, through their participation at the workshops, provided their contributions for our report. A full list of contributors can be found in the Appendix. Except for the direct quotes, the views and opinions expressed in this report represent the authors’ best efforts to synthesise the diverse perspectives shared by the contributors, and do not necessarily reflect those of individual contributors or reviewers.



3. The role of PGS data sharing in low-resource settings

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3.3	Constraints on PGS data sharing in low-resource settings	16
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3. The role of PGS data sharing in low-resource settings

Three key themes emerged from the workshops regarding the role of PGS data sharing in low-resource settings.

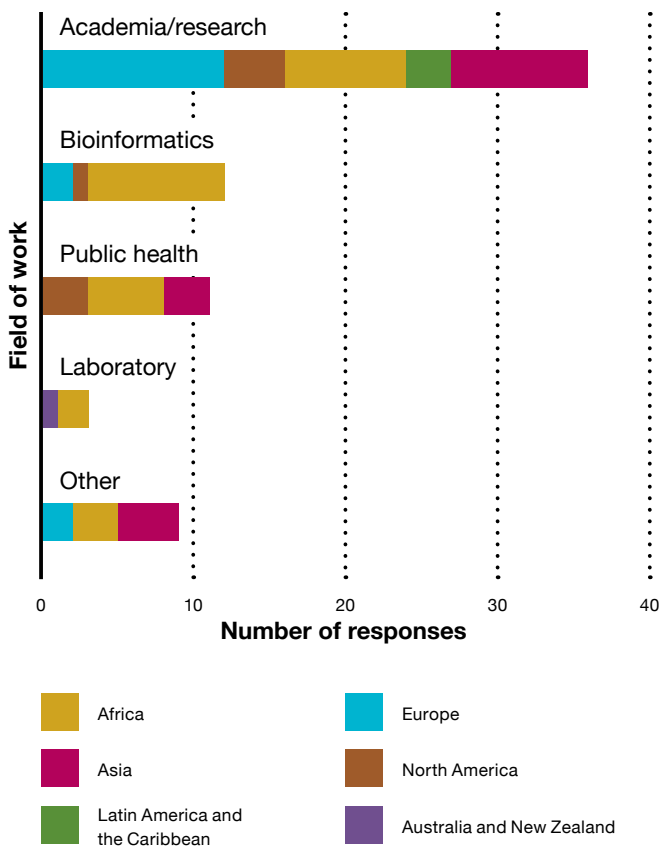
First, **collaboration as a lifeline** was repeatedly emphasised. In low-resource settings, PGS data sharing allows researchers to access global expertise and tools that they do not have locally. This collaboration is crucial for participation in global disease surveillance and pandemic preparedness efforts. Without access to international data sharing networks, these regions could struggle to keep up with pathogen evolution and global health responses. As a stakeholder from a low-resource setting noted, *“In low-resource settings, data sharing is the only way we can access the technologies and expertise we don’t have locally.”*

Second, **dependence on external resources** is a major theme and viewed as a double-edged sword. Local infrastructure often cannot support the tools needed for PGS data analysis, so researchers rely on platforms like [GISAID](#) and [GenBank](#), as well as regional alternatives. *“As an infrastructure provider... I’ve often seen data go unanalysed because of the lack of local infrastructure... We’ve seen it very clearly in South Africa, but there’s also perils of not sharing, and some of those perils are driven by lack of access to infrastructure, which is why we’ve looked at the use of non-local infrastructure,”* emphasised a workshop participant from South Africa.

Finally, **barriers to participation and attribution** were a consistent theme. Despite the clear need for PGS data sharing, many low resource settings face challenges such as limited internet connectivity, lack of funding, uncertainty over what data is appropriate to share and concerns over data misuse or misattribution. These issues prevent full participation in global data sharing platforms, limiting the contribution and impact of researchers in these regions. A workshop participant from a low-resource setting noted *“So many low-resource settings lack the necessary technological infrastructure, including reliable internet access and data storage solutions.”* Commenting on attribution, a workshop participant added, *“Although we did share data, what was not clear was who owns the data and there were instances where data was published, used, presented without acknowledging us.”*

3.1 Survey responses analysed by field of work and region

Wellcome shared a bespoke survey on understanding technical barriers to deposit, access and re-use of PGS data in low-resource settings in July 2024. This received 71 useable responses from stakeholders working with PGS data in both high-income (37%) and low- and middle-income (63%) settings. Most respondents generate and analyse PGS data for both research (77%) and public health purposes (79%), with respondents in low-resource settings most likely to be generating and analysing PGS data for disease surveillance (74%).



3.2 Use of technical solutions in different settings

Survey results from the global stakeholders, further corroborated by the workshops, showed that existing solutions for PGS data sharing and reuse can be categorised into three scenarios based on the availability of resources and capacity. Furthermore, these challenges impacted recognition, collaboration, and data reuse, in their settings, as in the table below.

Building on these findings, workshop participants further explored how these challenges impacted recognition, collaboration, and data reuse, in their settings.

3.2.1 The impact of technical solutions and infrastructure on recognition, collaboration and data re-use

	Minimal resources and capacity	Limited resources and capacity	Sufficient resources and capacity
Typical technical solutions	Tools such as email or physical devices such as external hard drives are often used. These methods provide a practical solution for data transfer when internet access or cloud infrastructure is limited.	Platforms like Nextcloud and WeTransfer are used for short-term sharing and transfer of PGS data. Nextcloud provides secure, self-hosted storage for collaboration, while WeTransfer allows quick transfers of large datasets. Platforms such as Agari and IBDC in India provide alternatives to regional needs.	Cloud-based platforms like Terra and CLIMB , supported by commercial cloud services, (AWS , Azure , or Google Cloud), or open-source tools like OpenStack . Open-source analysis tools like Galaxy , Bioconductor , and Nextflow provide data storage and management during workflows.
Digital infrastructure	Basic local IT infrastructure enables essential data sharing but is often not scalable or efficient . Cloud based systems may be inaccessible due to poor internet connectivity and limited computing power .	Solutions focus on simple, accessible tools that require minimal local infrastructure and allow secure, straightforward data exchange . Local and regional platforms provide a cost-effective way to share PGS data but are constrained by local infrastructures.	Cloud platforms offer scalable, cost-effective solutions for data storage, processing, and analysis without the need for extensive digital infrastructure.
Impact on recognition, collaboration and data re-use	Researchers may struggle to gain attribution due to informal nature of sharing. Limited opportunities to meaningfully collaborate as researchers cannot fully engage with global platforms, due to disparities in digital infrastructure.	Lack of support for complex cloud-based solutions , inhibits researchers' participation. Researchers can achieve some attribution when using regional platforms, but global recognition remains limited.	Researchers in these contexts are able to participate effectively in global data sharing initiatives. Collaboration is extensive with easy access to international partnerships and the ability to integrate data from multiple sources, for maximised data reuse.

3.3 Constraints on PGS data sharing in low-resource settings

Three critical challenges emerged from the survey: (1) infrastructure gaps, (2) limited expertise in using advanced bioinformatics tools, and (3) reliance on insecure data sharing methods.

The most significant **infrastructural** challenge noted across both high- and low-resource settings was the lack of proper interfaces to transfer data between local, national, and international systems, as well as from research environments to public health networks. In low-resource settings specifically, inadequate computational power, insufficient storage capacity, and unreliable internet connectivity pose significant barriers to PGS data sharing. Researchers often resort to using physical storage devices or low-security file-sharing services, which, in addition to the lack of security, are not scalable. As one survey respondent noted, *“Our infrastructure cannot handle the large datasets required for PGS data, so we often have to rely on physical storage devices or low-security file-sharing services.”* This limitation restricts participation in global data sharing initiatives and hinders recognition in the international research community.

Even when infrastructure is available, a lack of **expertise** in bioinformatics tools presents another challenge. Some institutions have sequencing equipment but lack personnel trained in advanced data analysis techniques. A bioinformatics specialist from West Africa commented, *“We have the machines, but no one to train us on the advanced tools needed to analyse this data properly.”* This expertise gap reduces engagement in high-impact research and limits opportunities for international collaboration.

Concerns over **data privacy and security** further complicate PGS data sharing in regions where advanced privacy-preserving functions are unavailable. Researchers may be reluctant to share sensitive data due to inadequate means of ensuring data security, especially with external partners. One respondent from a Southern African public health institute pointed out, *“We don’t have the means to ensure data security, so we’re cautious about sharing sensitive pathogen data, especially with external partners.”* This caution about sharing data in low-resource settings limits participation in global initiatives.



3.4 An enabling environment for data sharing

In the section that follows, actions that are focussed on technical solutions are presented first, in line with the workshops' primary focus. However, contributors emphasised that the challenges associated with data sharing are not solely technical in nature and a supportive 'enabling environment' is essential.



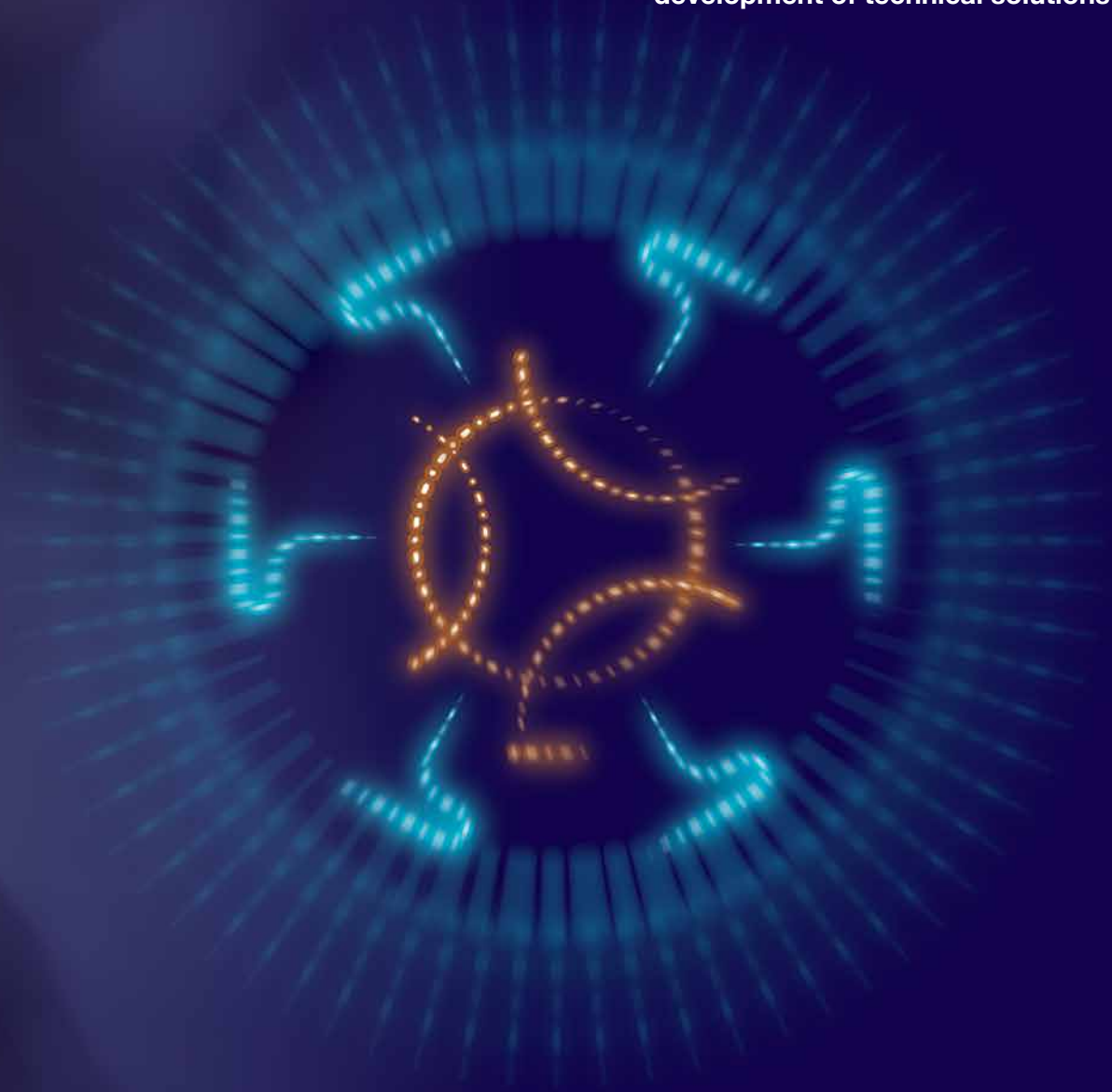
3.4.1 Characteristics of an enabling environment for PGS data sharing

Enabling theme	Enabling characteristics
Trust and Relationships	• Strong relationships are established between researchers, public health agencies, and policymakers. • Early engagement and co-creation processes inform the data that is gathered and shared. • Trust in data sharing systems has been established through user-friendly and fit-for-purpose design. • Effective collaborations are fostered at both technical and strategic levels, including policymakers.
Data Governance and Policy	• Clear data governance policies and licensing agreements are in place. • Alignment between institutional, national and funder policies enables cross-institutional data sharing. • Community best practices and FAIR data principles (that data are Findable Accessible Interoperable and Reusable) are promoted and followed. • Minimum metadata requirements have been defined for different data types.
Recognition and Credit	• Systems for proper recognition and crediting of data contributors are established. • Unique identifiers or digital object identifiers (DOIs) for datasets are consistently implemented. • Publication practices recognising and rewarding data sharing are adopted. • Acknowledgment of data contributors in research outputs is the norm.
Education and Advocacy	• Education programmes on data sharing benefits and processes are conducted, addressing fears around data sharing. • Training on data sharing best practices and standards is provided. • There is widespread promotion of success stories and case studies demonstrating the value of data sharing.



4. Technical solutions to enhance PGS data sharing

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4.2	Developing adaptable and context-specific solutions	22
4.3	Recommended approaches to the development of technical solutions	23



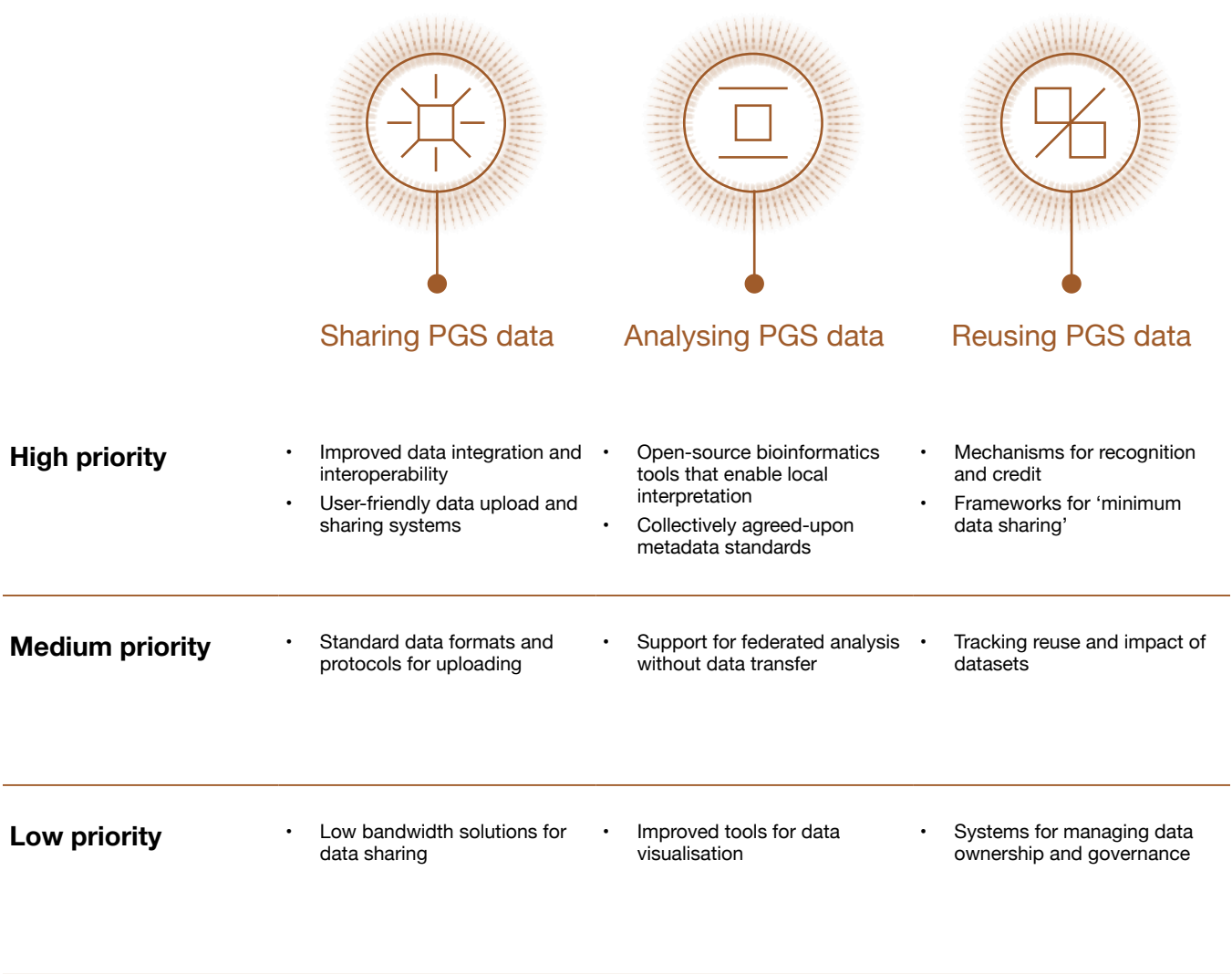
4. Technical solutions to enhance PGS data sharing

4.1 Strengthening technical solutions

Participants' proposals for strengthening existing technical solutions are grouped under the headings of sharing, analysing and reusing pathogen sequence data in the figure below. The suggestions were prioritised based on the frequency with which solutions were mentioned and the level of consensus on their significance.



4.1.1 Prioritisation of technical solutions for PGS data sharing





Sharing PGS data

Participants identified a need for more user-friendly data upload and sharing systems which are suitable for low-bandwidth contexts and stressed the value of standardised data formats and protocols in this context. *“One thing that we hear a lot is that we need ways to improve or simplify the process of putting data into databases. It really is a blockage. People can’t do it quickly, so they just don’t do it at all,”* noted one contributor.

To enable effective data exchange, solutions must also integrate with both local and international systems across platforms. An expert in data standards highlighted, *“We need to scale up rapidly in creating interoperable standards to allow data to flow freely between settings.”* Embedding these practices within local institutions can help to facilitate seamless collaboration across sectors.



Analysing PGS data

Open-source bioinformatics tools were highlighted as a key priority to enable data sharing without being bound to proprietary systems. One respondent emphasised, *“Open-source tools are critical because they allow low-resource settings to use and modify them without paying for licenses.”* Co-development with local stakeholders would help ensure these tools are adaptable to specific regional challenges.

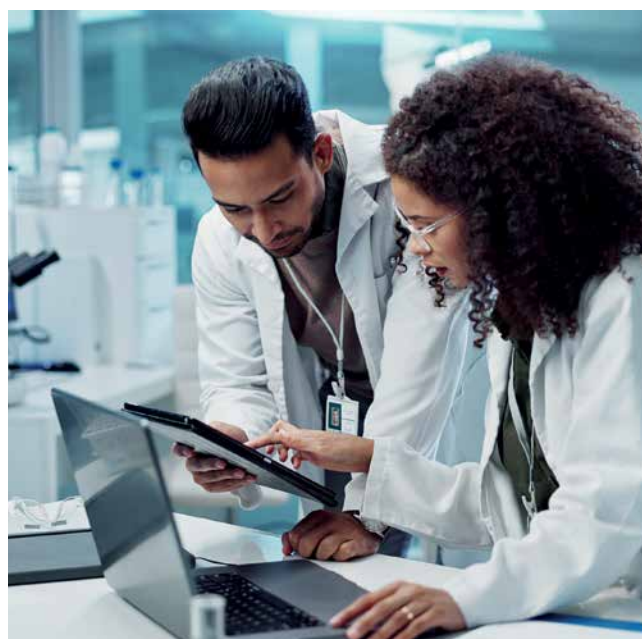
Support for data visualisation and federated analysis without the need for data transfer was noted as particularly impactful in low-resource settings. Some participants noted that developments made within the human genomic sequencing community could potentially be adapted for PGS data: *“The human data analysis community has a number of mechanisms for facilitating some degree of federated data analysis... This might be another area where there’s things that can be copied across.”*

By eliminating the necessity for extensive data transfer, federated data analysis not only conserves limited bandwidth but also maintains data privacy and sovereignty. Meanwhile, enhanced visualisation tools and collectively agreed-upon metadata standards can bridge the gap between data complexity and usability, making it easier for scientists in low-resource settings to interpret and act upon the data.



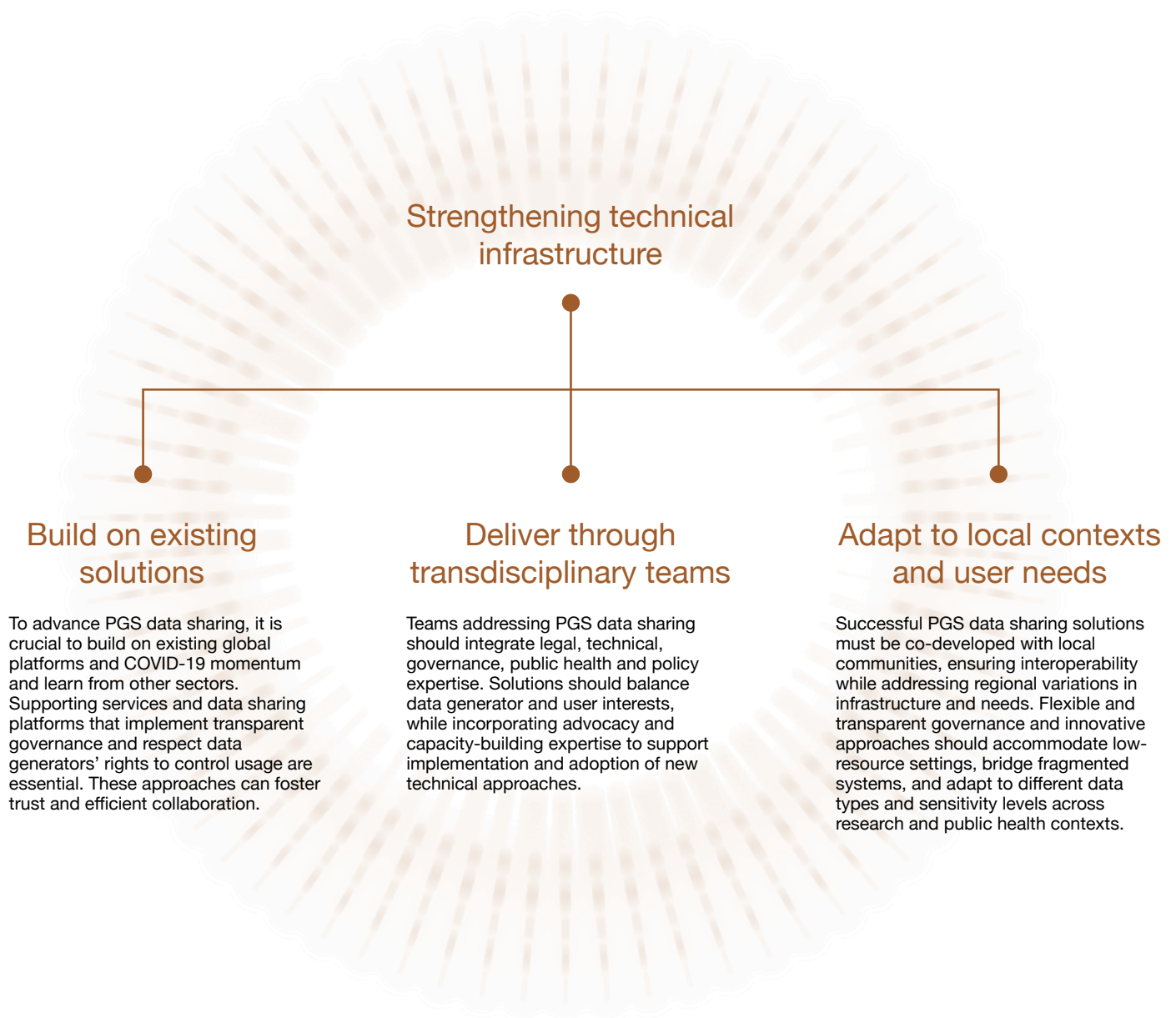
Reusing PGS data

Many of the challenges surrounding the sharing and reuse of PGS data are socio-cultural, political and legal in nature. One participant from a high-income setting argued that: *“The important thing is establishing community best practice, because technical solutions generally introduce more barriers and more silos, and they may benefit some people but not others.”* Technical solutions must be designed to address these concerns if they are to improve their utility and uptake by the scientific community. Solutions that prompt or require recognition and credit when reusing data help to foster a culture of openness and collaboration, for example, which can help ensure that contributors are acknowledged for their work. Establishing frameworks for ‘minimum data sharing’ and implementing these within relevant technical solutions helps standardise what is shared, lowering the barriers to reuse. Meanwhile, systems for managing data ownership, governance and licensing are crucial in defining rights and responsibilities, which can prevent misuse and ensure ethical standards are maintained. Additionally, tracking the reuse and impact of data sets provides valuable insights into the utility and reach of shared data, encouraging further participation and helping funders measure impact.



4.2 Developing adaptable and context-specific solutions

Workshop participants emphasised that successful technical solutions would need to be adaptable and context specific, ensuring interoperability and inclusion of local stakeholders. These additional considerations are summarised in the figure below, and further described in the remainder of this section.



4.3 Recommended approaches to the development of technical solutions

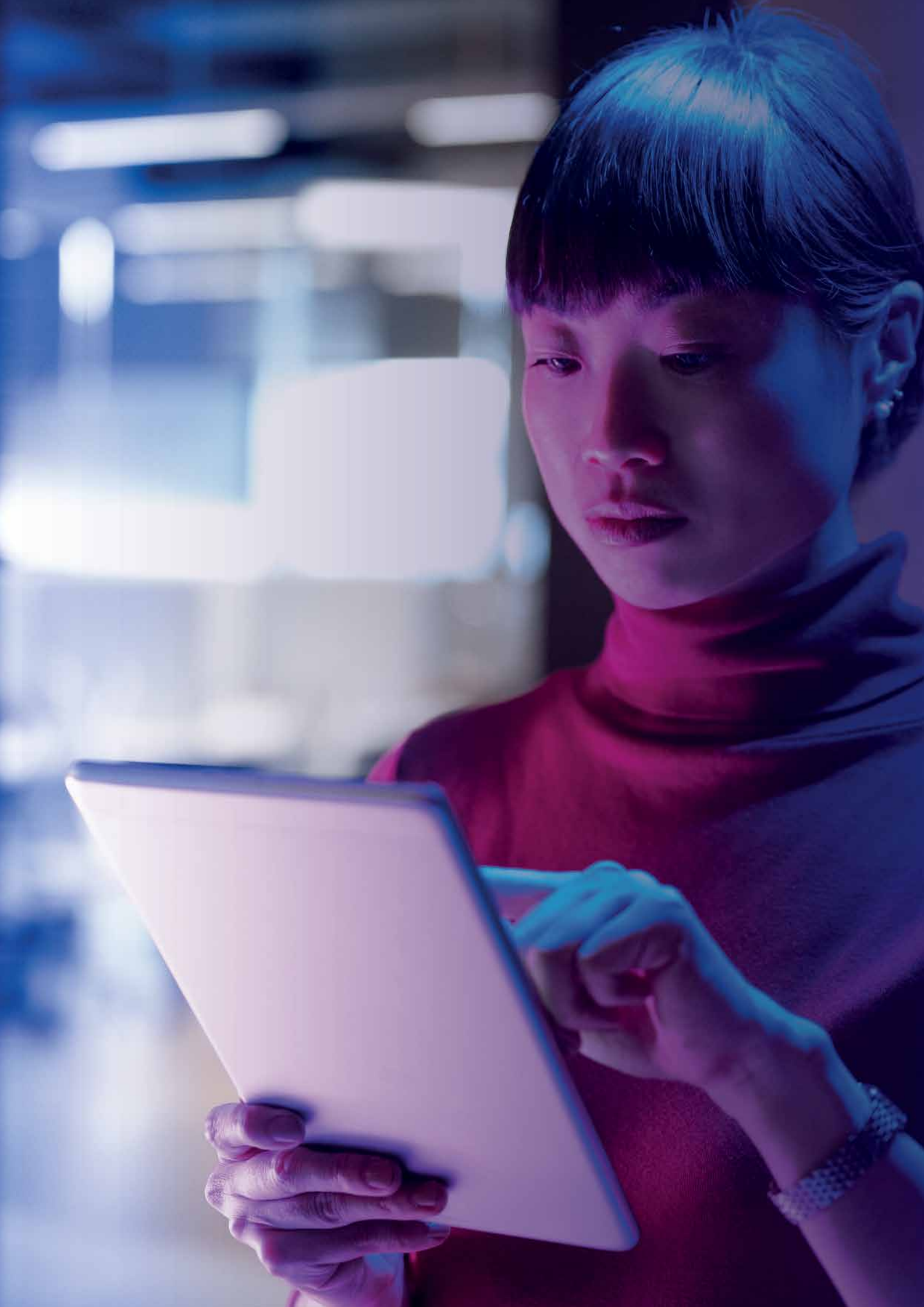
Workshop participants stressed the critical importance of new or improved technical solutions being adapted to specific local contexts and developed by transdisciplinary teams that include both academic and non-academic stakeholders¹. To achieve widespread adoption, technical solutions must also build on existing platforms, striking an appropriate balance between the requirements of data generators and users. Participants' recommendations under each of these themes are summarised in the table below.



4.3.1 Participants' recommendations for the development of technical solutions

Theme	Recommendations from participants
Build on existing solutions	• Leverage existing platforms that researchers have agreed upon globally, building on the momentum created during the COVID-19 pandemic for rapid data sharing. • Learn from successful data sharing implementations in other sectors, such as Africa's financial technology sector. • Adapt lessons from federated data analysis in human health to PGS data sharing. • Adopt and encourage funding of rapid, low-friction pathogen data sharing services. • Promote transparent governance structures for technical solutions to build trust in data sharing. • Address concerns about data ownership by giving data generators the opportunity to specify how their data can be used.
Deliver through transdisciplinary teams	• Assemble teams that combine expertise from the legal, technical, governance and policy sectors to tackle the intricate and multifaceted issues surrounding PGS data sharing. • Include public health actors alongside researchers to ensure data can be translated into action, recognising that these stakeholders' needs are distinct but overlapping. • Ensure the interests of both data generators and data users are considered in a balanced way as solutions are developed. • Include individuals with expertise in advocacy, training and capacity-building within the team, as these are integral to the development and uptake of new and improved technical solutions.
Adapt to local contexts and user needs	• Co-create solutions with local communities to ensure they are interoperable across systems and fit the local context. • Prioritise flexible governance structures and innovative data sharing solutions tailored to the specific needs of low-resource settings. • Recognise that regional needs vary; different countries may require distinct approaches based on their infrastructure, privacy concerns, and expertise. • Allow for the differing requirements of research versus public health. Solutions are needed to bridge fragmented systems within public health institutions and enable better connection between clinical, laboratory, and epidemiological data. • Adapt solutions and metadata standards in line with the nature and sensitivity of the data being shared.

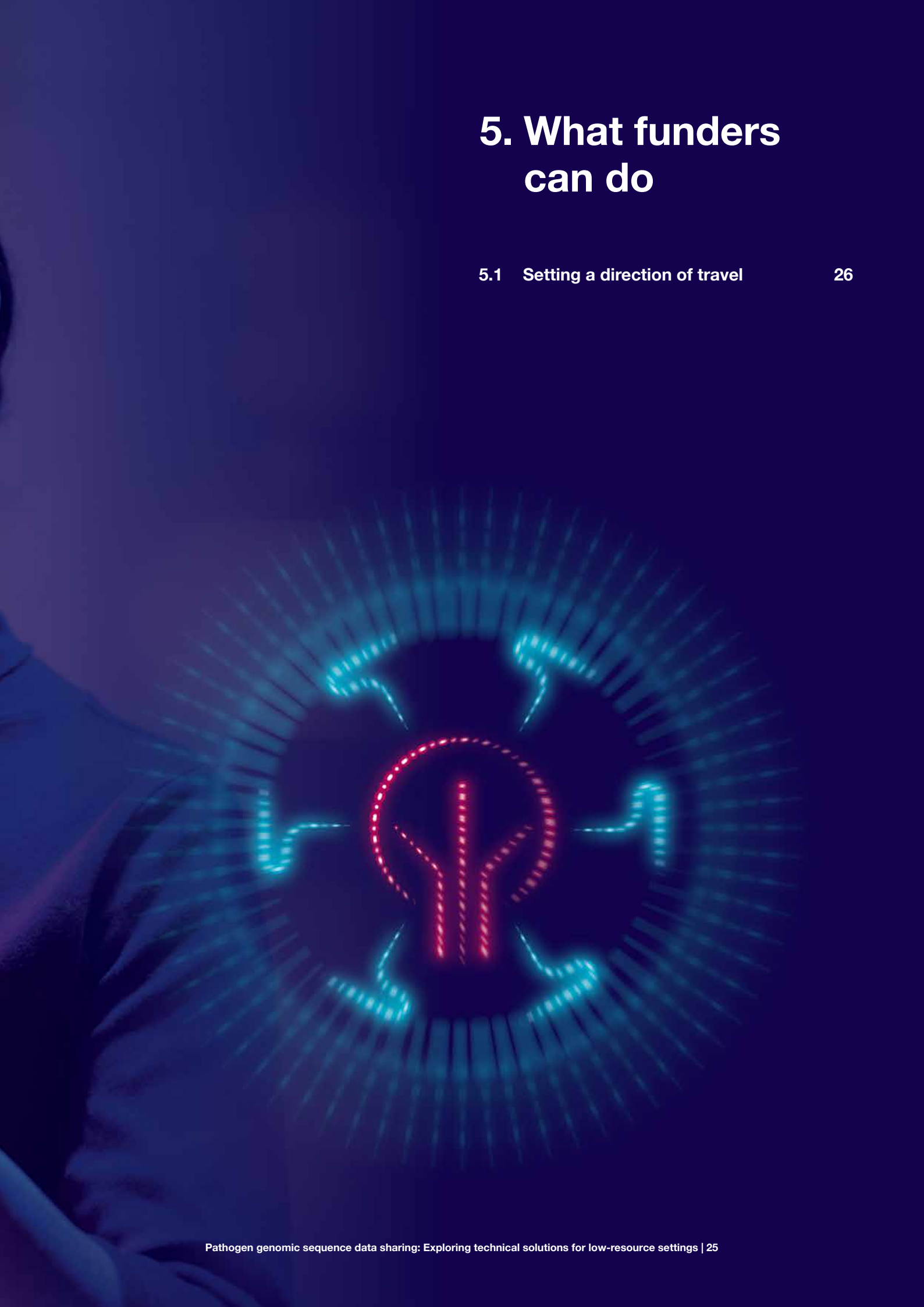
¹ The term 'transdisciplinary teams' is used in the sense of 'an approach that integrates knowledge across academic disciplines and involves collaboration with non-academic stakeholders to address societal challenges.' (Utrecht University, '[Transdisciplinary field guide](#)').



5. What funders can do

5.1 Setting a direction of travel

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5. What funders can do

5.1 Setting a direction of travel

The workshops showed that funders like Wellcome are in a strong position to drive improvements in global sharing of PGS data, particularly by those in low-resource settings. Participants highlighted the potential to enhance technical infrastructure for data upload, visualisation and analysis but stressed that interventions in this area must be accompanied by efforts to strengthen local capacity and build community norms. The interconnected nature of these activities points to a need for coordination between funding agencies to minimise duplication and maximise their collective impact.



5.1.1 How funders can support PGS data sharing in

low-resource settings

	Enhancing technical infrastructure	Strengthening local capacity	Building community norms
What funders can do	Support development of user-friendly data upload systems suitable for low-bandwidth contexts and interoperable with international systems.	Embed training in the development of new solutions, following a ‘train the trainer’ model for sustainable knowledge dissemination.	Promote adoption of globally applicable minimum standards and protocols for data collection and sharing.
	Support development of open source, federated data visualisation and analysis tools.	Tailor capacity-strengthening efforts to address both technical and contextual challenges.	Promote best practices for data sharing and recognition, addressing barriers in low-resource contexts.
Cross-cutting themes	Build on existing solutions and networks		
	Deliver through transdisciplinary teams		
	Adapt to local contexts and user needs		

Appendix

We would like to thank the following respondents for their participation at the workshops, as interviewees and/or their role in peer reviewing this report.

Name	Organisation and country	Workshop attendee	Interviewees	Peer reviewer
Alan Christoffels	South African National Bioinformatics Institute		✓	
Andrew Rambaut	University of Edinburgh, UK	✓		
Andy Yates	EMBL-European Bioinformatics Institute (EMBL-EBI)	✓		
Awadhesh Pandit	National Centre for Biological Sciences, India	✓		
Chaoren Chen	ETH Zurich	✓		
Chitra Pattabiraman	Independent Consultant (Pathogen Genomics), India	✓		
David Aanensen	Oxford Big Data Institute	✓		
Duncan MacCannell	Centers for Disease Control and Prevention, USA		✓	
Emma Griffiths	PHA4GE, Canada	✓		
Feng Zhu	Asia Pathogen Genomics Institute, Singapore	✓		
Gerald Mboowa	Africa Centre for Disease Control and Prevention	✓		
Isabella Oyier	KEMRI Wellcome, Kenya	✓		
Kasambula Arthur Shem	Ministry of Health, Uganda	✓		
Kathryn Holt	London School of Hygiene & Tropical Medicine	✓		
Masanori Arita	DNA Data Bank of Japan	✓		
Meera Chand	UK Health Security Agency	✓		✓
Neelika Malavige	University of Sri Jayewardenepura, Sri Lanka	✓		
Ndahafa Frans	Namibia Institute of Pathology	✓		
Nick Loman	University of Birmingham, UK	✓		
Nicky Mulder	Human Hereditary and Health in Africa		✓	
Nicki Tiffin	South African National Bioinformatics Institute	✓		
Peter van Heusden	South African National Bioinformatics Institute	✓		
Sandy Mak	Asia Pathogen Genomics Initiative, Singapore	✓		
Sarion Bowers	Wellcome Sanger Institute, UK	✓		
Simon Harris	Bill & Melinda Gates Foundation, UK	✓		
Tan Le Van	Oxford University Clinical Research Unit, Vietnam	✓		
ThankGod Ebenezer	EMBL-European Bioinformatics Institute (EMBL-EBI)	✓	✓	✓
Theo Sanderson	The Francis Crick Institute, UK	✓		
Tim Dallman	World Health Organisation	✓		
Tom Connor	Public Health Wales	✓		
Torsten Semmler	Robert Koch Institute, Germany	✓		



About Wellcome

Wellcome is a global charitable foundation which supports science to solve the urgent health issues facing everyone. We fund curiosity-driven research, and we're taking on three of the biggest health challenges facing humanity – climate change, infectious disease and mental health.

About Research Consulting

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